Research Proposal

Generative Discovery of Grain-Boundary Motifs in High-Entropy Alloys: A Closed-Loop AI System for Autonomous Microstructure Exploration

© Ishraque Zaman Borshon, PhD Candidate, Aerospace & Mechanical Engineering, University of Arizona, ishraque@arizona.edu

1. Introduction and Motivation

High-entropy alloys (HEAs) present a combinatorial design space spanning composition, atomic arrangement, and grain-boundary (GB) structure. Grain boundaries critically influence diffusion, mechanical strength, and stability, yet their systematic exploration remains intractable. Density functional theory (DFT) and experiments are too expensive for exhaustive search, while most machine learning approaches remain passive, learning only from pre-existing datasets.

Scientific discovery, however, is iterative: hypotheses are generated, tested, and refined under uncertainty. This proposal aims to build a closed-loop AI system that actively explores GB configuration space by generating candidate structures, evaluating their properties, and selecting the most informative next experiments.

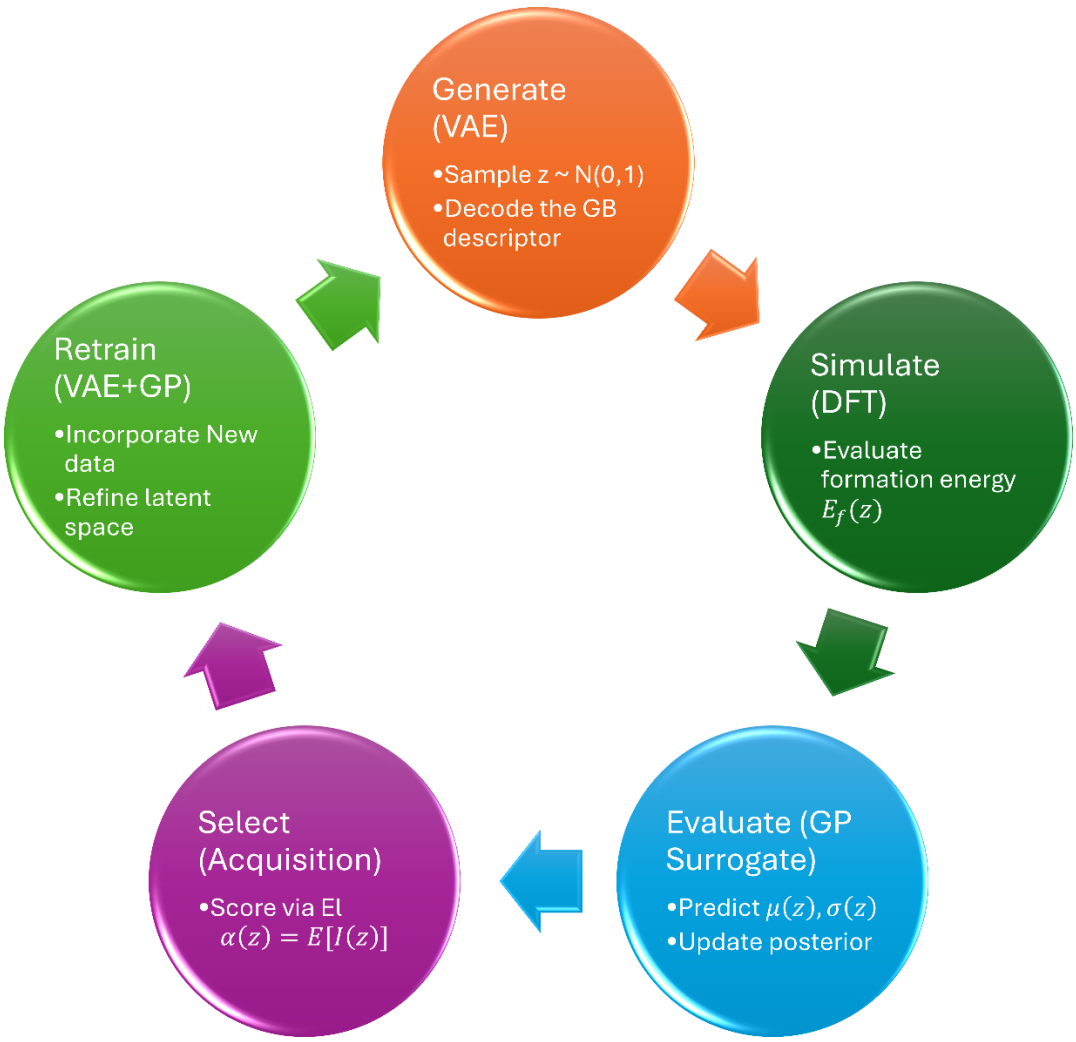

Figure 1: Proposed methodology for the Autonomous Microstructure Exploration

2. Prior Work and Feasibility

This proposal builds directly on pipelines I have already developed.

In my recent work on solid-electrolyte interphase (SEI) microstructures in lithium-ion batteries (Advanced Materials Interfaces, 2025), I developed an end-to-end framework integrating DFT-based structure generation, HRTEM simulation, and deep learning models with Bayesian uncertainty estimation. This system incorporated active learning to iteratively refine predictions under limited data.

In parallel, I have worked on high-entropy materials using both imaging and simulation. My work in npj Computational Materials (2024) applied deep learning to multicomponent oxide systems, while my recent study in Computational Materials Science (2026) combined HAADF-STEM, EDS, and DFT to analyze grain-boundary diffusion in Pt–Cu–Ir–Ni alloys, revealing element-specific transport pathways and activation barriers.

Together, these efforts demonstrate:

1. experience with grain-boundary physics in multicomponent systems,
2. development of physics-informed ML pipelines, and
3. use of simulation-in-the-loop learning frameworks.

This proposal extends these capabilities toward autonomous microstructure discovery in HEAs.

3. Core Idea

I propose a closed-loop generative AI system that explores grain-boundary configuration space using:

1. a generative model to propose candidate GB structures,
2. a surrogate model to estimate their energies and uncertainties, and
3. a Bayesian active learning strategy to select the most informative configurations for evaluation.

This system mimics the scientific method: generate hypotheses, evaluate them, and update knowledge iteratively. Unlike passive ML models, it actively navigates the search space toward physically meaningful and low-energy configurations.

4. Research Plan: Three Specific Aims

4.1. Aim 1: Build a Generative Latent Representation of HEA Grain-Boundary Structure Space

Objective: Train a generative model (VAE or equivariant generative network) on structural descriptors (e.g., SOAP, MBTR) derived from DFT-relaxed GB configurations. This model will learn a continuous latent space of physically valid grain-boundary motifs.

Deliverable: Ability to generate novel GB configurations beyond existing datasets.

4.2. Aim 2: Close the Loop with Bayesian Active Learning

Objective: Develop a Gaussian process surrogate model to predict GB formation energy and uncertainty from latent representations. Use Bayesian acquisition functions (e.g., Expected Improvement) to select candidate structures that maximize information gain.

Deliverable: Efficient exploration of GB space with minimal oracle queries.

4.3. Aim 3: Demonstrate Autonomous Discovery on a Target HEA System

Objective: Apply the closed-loop system to a representative HEA system (CrMnFeCoNi Cantor alloy, for which substantial DFT GB data exist) and quantify sample efficiency relative to random and grid-search baselines.

Deliverable: Benchmark results demonstrate that EI-guided active learning discovers low-energy GB configurations at 3-5x the sample efficiency of random search, with a curated library of novel candidate motifs prioritized for experimental validation.

5. Execution Plan

Months 1–2: Dataset assembly and descriptor generation; train generative model
Months 3–4: Implement active learning loop and surrogate model
Months 5–6: Benchmark and validate on HEA systems; release code and results
A working prototype of this pipeline has already been implemented using publicly available datasets and is available at:
https://github.com/ishraque2008/hea_gb_discovery

6. Alignment with Lila

This project directly aligns with Lila's vision of AI-driven scientific discovery:

1. Generative models: exploration of structure space
2. Bayesian methods: decision-making under uncertainty
3. ML-accelerated simulation: reducing reliance on expensive DFT
4. Agentic science: closed-loop hypothesis generation and testing

More broadly, this work treats AI not as a passive predictor but as an active participant in the scientific process, capable of guiding experiments and simulations toward high-impact discoveries.

References

Borshon, I. Z., et al. (2025). Deep learning analysis of solid-electrolyte interphase microstructures in lithium-ion batteries. Advanced Materials Interfaces. https://doi.org/10.1002/admi.202500558

Borshon, I. Z., et al. (2026). Evolution of grain boundaries in multi-principal alloys: Insights into phase stability and structural complexity. Computational Materials Science, 268, 114628. https://doi.org/10.1016/j.commatsci.2026.114628

Borshon, I. Z., et al. (2024). Predicting column heights and elemental composition in TEM images of high-entropy oxides using deep learning. npj Computational Materials. https://www.nature.com/articles/s41524-024-01461-w

Himanen, L., et al. (2020). DScribe: Library of descriptors for machine learning in materials science. Computer Physics Communications, 247, 106949. https://doi.org/10.1016/j.cpc.2019.106949

Batzner, S., et al. (2022). E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nature Communications, 13, 2453. https://doi.org/10.1038/s41467-022-29939-5

Batatia, I., et al. (2023). MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. Advances in Neural Information Processing Systems (NeurIPS 2022). https://arxiv.org/abs/2206.07697

Snoek, J., Larochelle, H., & Adams, R. P. (2012). Practical Bayesian optimization of machine learning algorithms. Advances in Neural Information Processing Systems (NeurIPS 2012). https://arxiv.org/abs/1206.2944

Kang, S., & Tamm, A. (2023). Density functional study of atomic arrangements in CrMnFeCoNi high-entropy alloy and their impact on vacancy formation energy and segregation. Computational Materials Science, 230, 112456. https://doi.org/10.1016/j.commatsci.2023.112456